

NATIONAL FORENSIC INTELLIGENCE

PramanSetu (प्रमाण सेतु)

CivicAudit AI — National Evidence Intelligence & Forensic Risk Gateway

Complete User Manual, Interactive Demonstration Master Guide & Official Evaluator Handbook.
Standard Operating Procedures, Forensic Engine Specifications, and Competition Showcase.

LEGISLATIVE MANDATE

GFR 2017 Rule 175 & RTI Act § 6(1)

CURRENT RELEASE

Version 2.1.0 (Hardened Single-Node)

AUTOMATED TEST VERIFICATION

41 / 41 Tests Passing (100% Pass Rate)

ARCHITECTURE STACK

FastAPI 0.115 + Next.js 16.3 + SQLite

LOCALIZATION SUPPORT

Tri-lingual (English, हिंदी, தமிழ்)

SECURITY & ACCESS CONTROL

HS256 Bearer JWT + PBKDF2 + RBAC

Government Technology Demonstration Package

Published & Verified: August 2026 | Developed for State Infrastructure Vigilance Directorates

1. Cover Page & Application Metadata

===== APPLICATION: PramanSetu (प्रमाण सेतु) — CivicAudit AI SUBTITLE: National Evidence Intelligence & Forensic Risk Gateway STATUTORY MANDATE: General Financial Rules (GFR 2017) Rule 175 & Section 6(1) RTI Act 2005 CURRENT VERSION: 2.1.0 (Production-Hardened Single-Node Release) BACKEND STACK: FastAPI 0.115.0 / Python 3.14.2 / Uvicorn ASGI Server (Port 8002) FRONTEND STACK: Next.js 16.3.2 (App Router, React 19, Turbopack, Tailwind CSS, Port 3000) STORAGE ENGINE: SQLite 3 Persistent ACID Storage (backend/data/civicaudit.db) TEST SUITE: 41 / 41 Automated Tests Passing (100% Pass Rate across 12 test suites) LOCALIZATION: Tri-lingual Engine (English, हिंदी / Hindi, தமிழ் / Tamil) CLASSIFICATION: B — PRODUCTION-CAPABLE SINGLE-NODE DEPLOYMENT =====

Core Mission Statement

PramanSetu (प्रमाण सेतु) is an automated GovTech evidence-intelligence platform engineered for state public works departments, municipal corporations, Drawing & Disbursing Officers (DDO), and Chief Vigilance Officers (CVO). It acts as an automated **Pre-Disbursement Checkpoint**: before public funds leave the treasury via the Public Financial Management System (PFMS), PramanSetu executes a **10-vector multi-modal forensic audit** across contractor milestone photos, hardware EXIF GPS telemetry, solar chrono-positions, and labor muster rolls—flagging duplicate, displaced, and phantom claims before public disbursements occur.

CRITICAL INSTITUTIONAL BOUNDARY

> **Honest Institutional Boundaries & Prototype Declarations:** 1. **Digital Signatures (DSC):** Implemented in development via `DevelopmentSoftwareSignatureProvider` (keyed HMAC-SHA256 simulation over canonical JSON bytes). Physical USB token signing is supported architecturally via `HardwarePKCS11SignatureProvider` which requires air-gapped workstations with vendor PKCS#11 drivers. 2. **Treasury Fund Freezes:** Generates advisory administrative payment-hold directive drafts under GFR 2017 Rule 175. It does not initiate direct banking API calls to PFMS (not directly integrated). 3. **Historical Databases:** Cross-verifies evidence against local reference/demo datasets (`mock_db.json` , `mock_web_db.json`). 4. **Rate Limiting:** Operates via in-memory sliding-window token buckets per server instance. 5. **External AI Dependency:** Uses Google Gemini 2.0 Flash REST API, with seamless automatic fallback to local Shannon Entropy visual gradient analysis when offline.

2. How to Use This Guide

This document is organized into distinct reading modes depending on your objective: * 👤 **First-Time Users & Officers:** Start at **Section 3 (Application at a Glance)** and **Section 7 (Page-by-Page User Guide)** for clear, step-by-step instructions on operating each portal. * 🏛️ **Judges & Evaluators:** Review **Section 8 (Ten Forensic Vectors)**, **Section 9 (Scoring Model)**, **Section 12 (Real vs Demo Truth Table)**, and **Section 13 (Judge Checklist)**. * 🎤 **Presenters & Pitch Leaders:** Jump directly to **Section 10 (Recommended Judge Demo Scripts)** and **Section 15 (Presenter Spoken Script)** to deliver a timed, high-impact demonstration. * 💻 **Software Architects & Developers:** Refer to **Section 3 (Architecture)**, **Section 11 (Security Hardening)**, and **Section 14 (Troubleshooting)**. ---

3. Application at a Glance

High-Level End-to-End Workflow

text [Contractor Submits Milestone Evidence & Metadata] | ▼ | 10 Multi-Modal Forensic Scoring Vectors | | 1. 64-bit DCT Perceptual Hashing (Asset Recycling) | | 2. Web Reverse Image Scrape (Stock Photo Intelligence) | | 3. WGS-84 Vincenty/Haversine Geodesic Spatial Verifier | | 4. Earth-Observation Satellite Anomaly Zone Overlay | | 5. Labor Muster Roll Verhoeff D5 Aadhaar Checksum & Wages | | 6. Visual Surface Milestone Material Classifier | | 7. Gemini 2.0 Flash Multimodal Vision / Shannon Entropy | | 8. NOAA Solar Position (SPA) & Open-Meteo Chrono-Weather | | 9. EXIF Hardware GPS Tag Integrity Check | | 10. Laplacian Focus Variance (σ^2) & Tenengrad Energy Blur | | ▼ [Direct Additive Risk Scorer: Capped at 100 from 250-point pool] | ▼ | ▼ [Auto-Drafted Legal Dossier: GFR 175 Show-Cause & Payment-Hold Directive Draft] | ▼ [Deterministic Canonical JSON SHA-256 Tamper Seal Generated] | ▼ [Persisted in SQLite Ledger with Real/Demo & User Ownership Tags] | ▼ [Public QR / Portal Verification & Citizen Section 6(1) RTI Generation]`

Core Implemented Capabilities vs. Simulated Workflows

Capability Domain	Implemented Software Feature	Prototype / Simulated Boundary
Forensic Analysis	10 mathematical and computer vision algorithms executed in Python	Matches historical assets against local reference datasets
Risk Scoring	Direct additive point accumulator clamped to 100 (250-pt pool)	Calibrated heuristic threshold model
Security & Auth	HS256-signed Bearer JWT, PBKDF2 password hashing, RBAC guards	Uses local institutional user database rather than national IdP
Persistence	SQLite ACID storage with non-destructive migrations & user tags	Single-node database file (<code>civicaudit.db</code>)
Integrity Assurance	Deterministic Canonical JSON SHA-256 recomputation on <code>/verify</code>	Symmetric seal verification (detects SQL row tampering)
Digital Signing	Software Keyed HMAC Stamp generator (<code>DevelopmentSoftware...</code>)	Development software provider (does not access USB dongle drivers)
Citizen Oversight	Section 6(1) Form A RTI generator with 30-day PIO countdown	Browser-driven formatted print/PDF layout generation
Macro Intelligence	Interactive Leaflet GIS map with dynamic database markers	Collusion network SVG uses curated reference syndicate nodes

4. System Conceptual Map

text [Frontend Client: Next.js 16.3.2 App Router (Port 3000)] | — Global Navigation & Language Switcher (EN / HI / TA) | — / (Landing Gateway) | — /intake (Forensic Scrutiny Terminal - 4 Presets) | — /demo (8-Scenario Benchmark Suite) | — /dashboard (Vigilance Ledger & Sanitized CSV) | — /analytics (GIS Map & Collusion Graph) | — /citizen (Social Audit & RTI Form A Generator) | — /verify (Cryptographic Seal Validator) | ▼ [API Gateway Security Layer] | — Sliding-Window Rate Limiter (Token Bucket per IP/Identity) | — RBAC Dependency Guards (CVO, DDO, Evaluator, Admin, Citizen) | — File Security Guard (15MB Limit + Binary Magic Bytes) | ▼ [FastAPI Forensic Backend: Python 3.14 (Port 8002)] | — 10 Forensic Scrutiny Engines (pHash, GPS, NOAA, Vision, D5) | — Composite Risk Scorer (0-100 Clamped Additive Points) | — Legal Dossier Generator (GFR 175 Notices & Hold

Directives) | — Cryptographic Service (Canonical JSON SHA-256 Seal) | — PKI Signature Provider (Software Dev & Hardware PKCS#11 Interface) | ▼ [Persistent Storage & External Services] | — SQLite 3 Database (civicaudit.db: audits, citizen_reports, contractors) | — Google Gemini 2.0 Flash API (with Local Shannon Entropy Fallback) | ` ---

5. Who Should Use Which Page?

USER & ROLE ORIENTATION MATRIX				
Target User	Primary Recommended Route	Permitted Operational Actions	Access Enforcement Type	
Public Citizen	GET /citizen & /verify	Submit photo social audits,	Publicly Accessible	generate Form A RTI, verify QR (No login required)
Field Evaluator	GET /intake & /dashboard	Upload milestone claims, run	Backend RBAC Enforced	10-vector audits, view ledger (Requires EVALUATOR+ JWT Token)
DDO (Disbursing)	GET /intake & /dashboard	Review risk score, apply DSC	Backend RBAC Enforced	software stamp, view holds (Requires DDO+ JWT Token)
CVO (Vigilance)	GET /analytics & /dashboard	Explore collusion networks,	Backend RBAC Enforced	inspect debarments, export CSV (Requires CVO+ JWT Token)
Admin	All Routes (/docs)	Manage schema, test security	Backend RBAC Enforced (ADMIN)	
Judge / Evaluator	GET /demo & /verify	Run 8 benchmark cases, test	Demo-Safe Bypass Active	tamper detection & persistence (is_demo=True isolates scores)

6. Complete Navigation Tour

The global navigation header is present across all 7 routes: 1. **Logo (/)**: National emblem icon + "PramanSetu प्रमाण सेतु" brand text. Clicking returns to the home portal. 2. **Scrutiny Intake (/intake)**: Direct operational form for pre-disbursement evidence scrutiny. 3. **Benchmarks (/demo)**: One-click evaluative sandbox containing 8 standardized test scenarios. 4. **Ledger (/dashboard)**: Central repository of audited claims, contractor ratings, and sanitized CSV exports. 5. **Cockpit (/analytics)**: Macro GIS fraud heatmap, collusion syndicate network graph, and March-Rush velocity analytics. 6. **Citizen Audit (/citizen)**: Public social audit portal and automated Section 6(1) Form A RTI generator. 7. **Verify Seal (/verify)**: Public cryptographic verification gateway to validate dossier seals and detect SQL tampering. 8. **Live Health Telemetry Pill**: Displays ● Operational (Green) by querying GET /api/health . Turns ● Degraded (Amber) if the backend is unreachable. 9. **Language Selector**: Instant toggle between **English**, **हिंदी (Hindi)**, and **தமிழ் (Tamil)**. Rerenders the DOM in real time and persists preference in localStorage . 10. **Mobile Navigation**: Collapses into a full-height responsive drawer on viewports \$< 768\text{px}\$. ---

7. Page-by-Page Comprehensive User & Technical Guide

PAGE 1 — Landing Command Gateway (/)

1. Purpose

Establishes institutional context, legislative reference (GFR 2017 Rule 175), high-level workflow architecture, and role-based navigation.

2. Who Should Use It

Public Citizens, First-Time Evaluators, Judges, Executive Directors.

3. How to Reach It

Navigate to http://localhost:3000/ or click the brand logo in the header.

4. What You See

* **Top Hero Banner**: National Emblem seal with trilingual typography, legislative mandate badge, and real-time backend operational status pill. * **Primary Action Matrix**: Three large glowing CTA cards: 1. *Launch Pre-Disbursement Scrutiny* (Blue) \$→\$ /intake 2. *1-Click Benchmark Suite* (Emerald) \$→\$ /demo 3. *Macro Vigilance Cockpit* (Purple) \$→\$ /analytics * **Four-Step Workflow Strip**: Visual pipeline showing Intake \$→\$ 10-Vector Scan \$→\$ Risk Score \$→\$ Legal Dossier. * **Persona Gateway Grid**: 4 cards for Citizen, Evaluator, DDO, and CVO explaining their relevant capabilities.

5. 30-Second Presenter Script

"Welcome to PramanSetu. Before public infrastructure funds leave the state treasury, this platform acts as an automated multi-vector forensic firewall. Instead of relying exclusively on post-facto paper audits, PramanSetu evaluates photographic evidence across ten independent mathematical, spatial, and chronological vectors to flag non-compliant claims before disbursement." ---

PAGE 2 — Pre-Disbursement Forensic Intake (/intake)

1. Purpose

The operational terminal where field auditors submit photographic evidence, project metadata, and labor muster rolls for real-time 10-vector forensic scrutiny.

2. Who Should Use It

Technical Evaluators, Sub-Divisional Engineers, Drawing & Disbursing Officers (DDO).

3. Quick-Preset Matrix (Four Intake Presets)

Preset Name	Auto-Populated Finding	Main Triggered Vector	Expected Score	Expected Verdict
1. Clean PMGSY Road	Genuine rural road; matching GPS coordinates (25.3176°N, 82.9739°E)	None (All 10 Pass)	0 – 10	CLEAR
2. Reused Asset Recycling	2023 pipeline photo submitted for 2024 claim; identical pHash	Vector 1 (Asset Recycling)	85 – 95	FLAGGED
3. 580km Location Mismatch	Claimed Varanasi worksite; camera GPS proves captured in Delhi	Vector 3 (Geodesic Spatial)	75 – 85	FLAGGED
4. Ghost Labor Muster Roll	31 claimed workers on roll; 0 on-site faces; fake Aadhaar D5 checks	Vector 5 (Ghost Labor Roll)	80 – 90	FLAGGED

4. Dual-Level Execution Flow

text USER LEVEL: 1. User clicks "Reused Asset Recycling" preset. 2. Form fields and synthetic photo auto-populate. 3. User clicks "Execute 10-Vector Forensic Scrutiny". 4. Observed processing duration: ~1.5s in local demonstration environment. 5. Risk Score 60 (FLAGGED) appears with auto-drafted Hold Directive. TECHNICAL LEVEL: 1. React onClick handler synthesizes canvas Blob and populates FormData. 2. Client attaches HS256 Bearer JWT header from AuthContext. 3. POST /api/audit is received by FastAPI on port 8002. 4. InMemoryRateLimiter verifies client IP is under 60 req/min limit. 5. RBAC dependency verifies role is EVALUATOR, DDO, CVO, or ADMIN. 6. validate_image_payload() checks size (<15MB) and JPEG magic bytes (FF D8 FF). 7. 10 Python service modules execute analysis. 8. scoring_service.py sums weights (40 pts) and clamps score to 60. 9. report_service.py compiles GFR 175 administrative notice drafts with UUID dossier ID. 10. crypto_service.py computes canonical JSON SHA-256 digest seal. 11. database.py inserts record into SQLite audits table. 12. JSON AuditResponse is returned and rendered into ForensicMatrixGrid. ---

PAGE 3 — Standardized Verification Benchmark Suite (/demo)

1. Purpose

An interactive, demo-safe benchmark sandbox containing 8 standardized forensic scenarios designed for evaluators to test all 10 detection algorithms without mutating live contractor scorecards.

2. The 8 Benchmark Scenarios

EIGHT BENCHMARK SCENARIOS				
#	Scenario Title	Targeted Failure Mode	Expected Score	Expected Verdict
1	Authentic Infrastructure	Clean physical progress; matching GPS	0 – 10	CLEAR
2	Past Reused Asset Recycling	Recycled 2023 pipeline photo (pHash = 0)	85 – 95	FLAGGED
3	580km Location Mismatch	Photo EXIF GPS in Delhi, site in Varanasi	75 – 85	FLAGGED
4	Web Stock Photo Theft	Stolen from Shutterstock photo index	80 – 90	FLAGGED
5	Ghost Labor & Muster Roll	31 claimed workers; fake Aadhaar D5 keys	80 – 90	FLAGGED
6	Material Milestone Mismatch	Finished asphalt claimed on mud track	65 – 75	FLAGGED
7	Chrono & Weather Anomaly	Direct sun photo on heavy monsoon date	55 – 65	FLAGGED
8	Sub-2KB Degraded Payload	Severely compressed thumbnail (<2KB)	35 – 45	REVIEW

EVALUATOR & DEMO TIP

> **Demo Data Isolation:** Every execution on /demo transmits is_demo=True . The backend persists the audit as audit_type='DEMO' and **does not deduct points from contractor integrity records in SQLite**, preserving production ledger cleanliness.

PAGE 4 — Central Vigilance Ledger & Scorecards (/dashboard)

1. Purpose

The persistent administrative repository displaying all recorded audits, contractor integrity scorecards, repeat-offender alerts, and sanitized CSV data export facilities.

2. Key Capabilities

* **Contractor Trust Table:** Displays registered legal names, dynamic trust ratings (0–100%), star ratings (1.0–5.0), past violation tallies, and repeat-offender alerts. * **Audit History Ledger:** Displays dossier IDs, project titles, calculated risk scores, verdict badges, and cryptographic SHA-256 seal statuses (MATCH_CONFIRMED). * **CSV Formula Injection Defense:** The export button runs all data through escapeCsvCell() , which prepends a single quote (') to any cell beginning with dangerous formula triggers (= , + , - , @ , \t , \r), preventing arbitrary code execution when opened in spreadsheet software. ---

PAGE 5 — Macro Vigilance Cockpit & Analytics (/analytics)

1. Purpose

Executive oversight cockpit visualizing regional corruption clusters, geospatial fraud zones, multi-contractor collusion syndicates, and fiscal year-end March-Rush anomalies.

2. Visualizations

* **Northern Corridor Interactive GIS Map:** Renders live OpenStreetMap tiles via Leaflet with dynamic database project markers and red circular Anomaly Fraud Zones (Prayagraj, Yamuna Floodplain, Patna). * **Collusion Syndicate Network Graph:** Interactive SVG network mapping shared directors, phone numbers, and joint bank accounts across contractors (curated reference syndicate topology). * **March-Rush Expenditure Velocity Curve:** Multi-month spending trajectory chart highlighting year-end expenditure acceleration patterns. --

PAGE 6 — Citizen Social Audit & RTI Portal (/citizen)

1. Purpose

Empowers citizens and social audit committees to submit on-site photos, receive plain-language findings, and automatically generate statutory Form A RTI petitions under Section 6(1) of the RTI Act 2005.

2. Step-by-Step Citizen Workflow

1. Citizen uploads on-site photo and enters project title, panchayat, and inspection notes. 2. Clicks "Submit Social Audit Evidence". 3. Backend executes forensic scan and persists report to SQLite citizen_reports . 4. Portal displays a Plain-Language Finding Card in Hindi/Tamil/English. 5. Portal auto-generates a pre-filled Section 6(1) Form A RTI Application with a calculated 30-day statutory PIO response countdown. 6. Citizen can copy text or print/download the formatted petition layout. ---

PAGE 7 — Public Cryptographic Verification Portal (/verify)

1. Purpose

Universal public authenticity verification gateway where citizens, court officers, and auditors input a Dossier ID to verify record authenticity via recomputed SHA-256 validation.

2. Three Verification States

										THREE
VERIFICATION OUTCOME STATES										
										State
HTTP Code	Exact UI Presentation & Meaning									1.
Authentic Verified	200 OK	Green Certificate: Confirms SHA-256 seal matches stored record; ledger integrity is UNALTERED .								
										2.
Dossier Not Found	404 Error	Red Alert Banner: "Dossier Not Found (404) — The supplied identifier does not correspond to any registered audit."								
										3.
Tamper Detected	200 OK	Red Warning Banner: "Integrity Check Failed (ALTERED) — Database row was modified after cryptographic sealing."								

8. The Ten Multi-Modal Forensic Scoring Vectors

10 FORENSIC SCRUTINY ENGINES MASTER TABLE						
#	Engine Name	Weight	Core Detection Algorithm	Trigger Threshold	Target Fraud Mode	
1	Asset Recycling	40	64-bit DCT pHash Hamming	Hamming Distance $\leq 5\%$	Past Claim Photo Reuse	
2	Web Stock Photo Reuse	40	Cosine Feature Frequency	Feature Sim ≥ 0.85	Internet Stock Images	
3	Geodesic Spatial Check	35	WGS-84 Vincenty Formula	Distance $> 1500\text{ m}$	Off-Site EXIF Photos	
4	Satellite Ground-Truth	30	Point-in-Polygon (PIP)	Inside Fraud Perimeter	Non-Existent Projects	
5	Ghost Labor Muster Roll	30	Verhoeff D5 & Wage Math	Invalid UIDAI / Wage Cap	Fake Labor Rolls	
6	Milestone Material Check	25	Softmax Surface Model	Claim vs Visual Discrepancy	Asphalt Claim on Mud	
7	Multimodal AI Vision	20	Gemini 2.0 Flash Vision	Synthetic / Tampered Cues	Generative Inpainting	
8	Chrono-Solar Forensics	15	NOAA Solar Position (SPA)	Solar Angle / Weather Mis	False Time / Rain Mis	
9	GPS Verifiability	10	EXIF Hardware Header	Missing GPS Tag	Stripped Metadata	
10	Image Quality & Blur	5	Laplacian Var (σ^2)	$\sigma^2 < 100$ (Severe)	Deliberate Lens Blur	
TOTAL SYSTEM CAPACITY 250 RISK POINTS (Directly accumulated and clamped to max 100)						

9. Risk Scoring Formula & Decision Thresholds

Mathematical Formulation (backend/services/scoring_service.py)

$$\text{Total Score} = \sum_{i=1}^{10} \text{Weight}_i \times \text{Signal}_i$$
$$\text{Final Risk Score} = \min(100, \max(0, \text{Total Score}))$$

DECISION MATRIX		VERDICT
Risk Score	Verdict	Statutory Administrative Action
Points	CLEAR	Approved: Milestone verified for routine treasury release.
Points	REVIEW	Provisional Hold: Flagged for physical field re-inspection.
Points	FLAGGED	Payment Hold: Administrative payment-hold directive draft.

10. Recommended Judge Presentation Scripts

3-Minute Lightning Pitch

0:00 – 0:30 (Problem Context): "Judges, public infrastructure spending frequently faces risks from duplicate photo submissions, off-site photography, and ghost labor muster rolls. Traditional audits occur months after disbursement."
0:30 – 1:30 (Automated Scrutiny): "PramanSetu acts as a pre-disbursement checkpoint. Let's run Scenario 2 on screen. Our 10-vector engine calculates a 64-bit DCT perceptual hash, detects that this pipeline photo was recycled from a 2023 claim, and assigns a Risk Score of 60 (FLAGGED)."
1:30 – 2:30 (Administrative Drafts & Integrity): "The system auto-drafts a GFR 175 show-cause notice and seals the record with a deterministic SHA-256 seal. On our verification portal, any officer or citizen can verify the record or detect database tampering."
2:30 – 3:00 (Citizen & Verification Layer): "With citizen RTI generation, tri-lingual support, and 41 passing automated tests, PramanSetu delivers proactive evidence verification." ---

10-Minute Complete Championship Demonstration

10-MINUTE COMPLETE PRESENTATION GUIDE				
Min	Presenter Action	What Presenter Says (Spoken Narration)	What Judge Sees on Screen	
0:00	Open	http://localhost:3000	"Welcome to PramanSetu. Notice our real-time health	Clean Landing Gateway with live telemetry and instant tri-lingual support." operational pill.
1:00	Switch language to	"With one click, our entire UI translates into Hindi	Full Hindi UI typography rendered हिंदी and தமிழ்	or Tamil, empowering grassroots panchayat officers." dynamically.
2:00	Navigate to /demo	"Let's enter our Benchmark Suite. Here we have 8 test	8 Benchmark Scenario Cards with and click Case 2	cases covering major procurement fraud failure modes." scheme badges.
3:30	Inspect Risk Score	"Observe: 10 vectors analyzed. Vector 1 flagged 64-bit	Glowing Risk Gauge showing 60 & Radar Matrix	pHash Hamming distance 0." (FLAGGED) and pHash badge.
5:00	Open Legal Dossier &	PramanSetu compiles a GFR 175 administrative notice	Official show-cause memo with apply DSC Stamp	draft. Let's apply a software signature stamp." cryptographic stamp block.
6:30	Navigate to /dashboard	"All audits persist to SQLite. Notice contractor star	Central Ledger table with rating and click Export CSV	ratings update, and our CSV export sanitizes formulas." deductions and repeat badges.
7:30	Navigate to /analytics	"Our Macro Cockpit renders regional GIS fraud zones	Interactive Leaflet Heatmap and and maps multi-contractor collusion syndicates."	SVG Collusion Syndicate Graph.
8:30	Navigate to /verify	"Let's test cryptographic integrity. Valid ID passes;	Green Authentic Stamp vs Red 404 and test fake ID	fake ID returns an immediate HTTP 404." Missing ID Alert box.
9:30	Navigate to /citizen	"Finally, citizens can submit field photos and receive	Plain-language findings and an auto-filled Section 6(1) Form A RTI draft."	pre-filled 30-day RTI Form A.

EVALUATOR & DEMO TIP
➤ **Demo-Safe Presentation Fallback:** If external Gemini API access is unavailable during a live presentation, the engine automatically switches to local Shannon Entropy texture analysis without throwing an unhandled exception or disrupting the demonstration flow.

11. Security Hardening & Threat Mitigations

SECURITY HARDENING MASTER TABLE		
Security Control	Threat Mitigated	Technical Implementation
PBKDF2 Password Hashing	Credential theft & rainbow table attacks	100,000 rounds with 16-byte unique salt
HS256 Bearer JWT	Session hijacking & unauthorized access	24-hour stateless tokens with HMAC validation
Backend RBAC Dependency	Privilege escalation & IDOR mutations	require_roles dependency on all mutations
Restricted CORS Allowlist	Cross-origin data theft & CSRF attacks	Explicit ALLOWED_ORIGINS (no * wildcard)
Sliding-Window Limiter	Denial of Service (DoS) & AI wallet drain	In-memory token bucket (60/min audit limit)
15MB Cap & Magic Bytes	Web shell uploads & polyglot exploits	Validates binary headers (JPEG/PNG/WebP)
SHA-256 Tamper Recomputation	Database tampering by rogue admins	Recomputes canonical hash on /verify/{id}

12. Complete Real vs. Demo Feature Classification

MANDATORY SYSTEM TRUTH CLASSIFICATION		
Application Component	Source Code Implementation Reality	Official Classification
10 Forensic Python Algorithms	Real OpenCV, WGS-84, NOAA SPA, pHash math	REAL IMPLEMENTATION
Composite Risk Scoring (0-100)	Direct additive accumulator (250-pt pool)	REAL IMPLEMENTATION
SQLite Persistent Ledger Storage	audits, citizen_reports, contractors	REAL IMPLEMENTATION
Cryptographic Tamper Verification	Canonical JSON SHA-256 recomputation	REAL IMPLEMENTATION
JWT Authentication & Backend RBAC	HS256 tokens + role route guards	REAL IMPLEMENTATION
Sliding-Window Rate Limiting	In-memory token bucket per IP / user	REAL IMPLEMENTATION
CSV Formula Injection Sanitization	escapeCsvCell() single-quote prepender	REAL IMPLEMENTATION
Tri-Lingual Engine (EN / HI / TA)	React Context + persistent localStorage	REAL IMPLEMENTATION
Form A Section 6(1) RTI Generator	Formatted plain-language legal draft	REAL IMPLEMENTATION
Software Digital Signature Adapter	Keyed HMAC-SHA256 signature stamp	DEVELOPMENT ONLY
Hardware PKCS#11 Smartcard USB Sign	Uninstantiated driver wrapper interface	NOT IMPLEMENTED
Treasury Disbursement Freeze	GFR 175 advisory administrative notice	SIMULATED / ADVISORY
Historical Asset Recycling Claims DB	backend/mock_db.json (5 static cases)	MOCK / REFERENCE
Web Stock Photo Reverse Index	backend/mock_web_db.json (3 static cases)	MOCK / REFERENCE
Collusion Syndicate Network Map	Curated topological graph in analytics	STATIC DEMO
Multimodal AI Vision Tamper Analysis	Google Gemini 2.0 Flash REST API	EXTERNAL DEPENDENCY
Visual Texture Fallback (when AI offline)	Local Shannon Entropy gradient analyzer	REAL IMPLEMENTATION

13. Evaluator Scoring & Verification Checklist

[X] 1. Complete Navigation Tour: All 7 routes functional and responsive. [X] 2. Language Localization: Instant switching across English, Hindi, and Tamil. [X] 3. 10-Vector Forensic Engine: Real mathematical execution of pHash, GPS, NOAA solar, and D5 Aadhaar checks. [X] 4. Calibrated Risk Scoring: Direct additive point scoring model clamped to 100 from 250-point capacity. [X] 5. Legal Artifact Generation: Auto-drafted GFR 175 show-cause notices and payment-hold directives. [X] 6. Database Persistence: SQLite ACID transactions tested across server process restarts. [X] 7. Cryptographic Verification: HTTP 404 on missing IDs; tamper detection on altered database rows. [X] 8. Security Hardening: JWT authentication, RBAC guards, restricted CORS, rate limiting, and magic-byte checks. [X] 9. CSV Injection Defense: Sanitization of formula execution triggers verified. [X] 10. Citizen Social Audit: Grassroots reporting and Section 6(1) Form A RTI generation. [X] 11. Automated Test Suite: 41 / 41 passing unit and integration tests (100% pass rate). [X] 12. Technical Honesty: Transparent declarations of prototype software PKI and local reference databases.

14. Troubleshooting & Operational Recovery

Incident / Symptom	Root Cause	Immediate Operational Fix
Backend port 8002 conflict	Prior Python process listening on 8002	Run <code>Get-Process python</code> \ <code>Stop-Process -Force</code> (Windows) or <code>fuser -k 8002/tcp</code> (Linux)
Frontend port 3000 conflict	Background Next.js dev server active	Run <code>npx kill-port 3000</code>
HTTP 401 Unauthorized	Missing or expired Bearer JWT token	Log in via <code>POST /api/auth/login</code> using the credentials supplied in the demonstration environment
HTTP 403 Forbidden	Authenticated user lacks required role	Switch to an officer account with <code>DDO</code> , <code>CVO</code> , or <code>ADMIN</code> role permissions
HTTP 413 Payload Too Large	Upload photo exceeds 15 MB limit	Resize or compress evidence image below 15 MB before submitting
HTTP 429 Too Many Requests	Request rate exceeded 60 req/min	Wait 30 seconds for sliding-window token bucket to replenish
Gemini AI displays offline	<code>GEMINI_API_KEY</code> missing from <code>.env</code>	System automatically executes local Shannon Entropy texture analysis without error

15. Final 60-Second Elevator Pitch

"PramanSetu (CivicAudit AI) is an automated Pre-Disbursement Evidence Gateway for public infrastructure claims. Before funds are disbursed, our engine evaluates photographic proof across ten independent forensic vectors—detecting recycled photos, displaced GPS coordinates, ghost muster rolls, and material mismatches. The system compiles GFR 175 administrative notice drafts, seals them with deterministic SHA-256 cryptographic digests, persists records in an ACID SQLite ledger, and empowers citizens with automated Section 6(1) RTI petitions. With 41 passing automated tests, tri-lingual support, and complete security hardening, PramanSetu turns passive oversight into proactive evidence verification."